

Effect of temperature on growth, metabolism, and gas exchange in *Hermetia illucens* larvae reared under commercial and laboratory conditions

M.L. Schøn^{1,2}, M.V.N. Mikkelsen³, K. Jensen¹, J.M. Poulsen¹, I.E. Berggreen¹, T.M. Schou⁴, J.V. Nørgaard¹, and J. Overgaard^{3,*}

¹ Department of Animal and Veterinary Sciences, Aarhus University, Blichers Alle 20, 8830 Tjele, Denmark

² Sustainable Food Processing Laboratory, Department for Health Sciences and Technology, ETH Zurich, Schmelzbergstrasse 9, 8092 Zurich, Switzerland

³ Zoophysiology, Department of Biology, Aarhus University, C.F. Møllers Allé 3, 8000 Aarhus C, Denmark

⁴ ENORM Biofactory A/S, Hedelundvej 15, 8762 Flemming, Denmark

*Corresponding author: Johannes.overgaard@bio.au.dk

^a See ORCID ID in the footnote

Running header: Metabolism and growth of BSFL under commercial and laboratory conditions

Abstract

The world's socio-economic development is continuously increasing the demand for efficient production of food, feed, and energy from the agricultural sector. In this respect, the emerging production of black soldier fly larvae (BSFL) represents a promising system to upscale low-quality resources to higher-quality resources usable as feed or biodiesel. To optimize BSFL production it is critical to establish methods that are relevant to examine how rearing conditions, including temperature, affect growth, metabolism, and body composition at large scale. In a first set of experiments, we therefore compared how small-scale laboratory respirometry chambers (600 larvae) support gas exchange, growth, and metabolism in black soldier fly larvae (BSFL) with similar measurements obtained from large-scale commercial conditions (>12 million larvae) at ~34-35°C. Having observed that small- and large-scale systems generated comparable measurements of growth and metabolism, we used our small-scale respirometry system to examine in detail how rearing temperature ranging from 27.0 to 41.9°C influenced growth, metabolism and body composition in BSFL. Here we found that 7-day growth of BSFL was stable and high at 27, 35, and 39°C, while a rearing temperature of ~42°C causes a severe, but sublethal, depression of growth and metabolism. Despite the general similarities in total growth over the 7-day period at 27, 35, and 39°C, we find considerable acceleration of metabolism and development at the two highest rearing temperatures (35 and 39°C) while lowered rearing temperature (27°C) reduces CO₂ production per kg larvae produced and the lower temperature is also associated with a higher crude protein to crude lipid ratio in larvae after the 7-day growth period. Based on these findings we discuss how continued monitoring of gas exchange in production systems holds a potential to optimize production in the emerging industry of large-scale commercial insect production.

Keywords: Oxygen consumption rate, CO₂ production rate, Respiratory exchange ratio, Black soldier fly larvae, Protein content, Lipid content

^a Maja Lermack Schøn,
Kim Jensen,
Ida Elisabeth Berggreen,
Toke Munk Schou,
Jan Værum Nørgaard,
Johannes Overgaard,

<https://orcid.org/0009-0007-6251-9034>
<https://orcid.org/0000-0003-0261-3831>
<https://orcid.org/0000-0002-8546-6517>
<https://orcid.org/0000-0001-7545-7572>
<https://orcid.org/0000-0002-0580-0184>
<https://orcid.org/0000-0002-2851-4551>

Introduction

The continued growth of the human population combined with an expected improvement in global socio-economic conditions will raise the demand for increased agricultural production (Alexandratos and Bruinsma, 2012; Tilman et al., 2011). However, agriculture is a major contributor to climate change through emissions of greenhouse gases (GHG), and agricultural land use has severe consequences for biodiversity (Campbell et al., 2017; Foley et al., 2011). These integrated and intertwined problems challenge the agricultural sector to promote and develop production systems that increase yields, reduce CO₂ emissions, improve energy management, and optimize the recycling of resources. Insect production, and in particular the production of black soldier fly (*Hermetia illucens*) larvae (BSFL), has been promoted as an industry with the potential to mitigate some of these challenges as it has the potential to recycle or upcycle low-quality organic matter to support the production of food, feed or biodiesel (Cortes Ortiz et al., 2016; Gahukar, 2016; Tabassum et al., 2016). Compared to traditional livestock production, the production of insects emits less GHG, requires less land, and utilizes less water (Gahukar, 2016; Oonincx and de Boer, 2012; Tabassum et al., 2016). Furthermore, this generalist detritivorous insect (Danieli et al., 2019) can bio-convert nutrients from organic waste into larval protein and lipid, which can be used as a supplement in livestock feed (Abd El-Hack et al., 2020; Barragan-Fonseca et al., 2017), or in the case of non-feed grade biomasses, as biodiesel and other non-food products depending on national regulations (Kaczor et al., 2023; van Huis, 2022; Zheng et al., 2012). Insect production is also characterized by high feed conversion efficiency and high productivity (Gahukar, 2016). Even so, all production systems, including BSFL production, can be optimized through an improved understanding of the abiotic and biotic factors that influence development, growth, and metabolism (Gillott, 2005).

Temperature is one of the most important factors influencing the survival, developmental rate, growth, and metabolism of insects (Neven, 2000; Shumo et al., 2019; Jørgensen et al., 2021; Ørsted et al. 2022; Chia et al., 2018). Consequently, rearing temperature also influences feed conversion efficiency (Gligorescu et al., 2018), as well as the proportional emission of GHG, and some studies have indicated that rearing temperature influences the product quality of the insects in terms of lipid-to-protein ratio (Bjørge et al., 2018; van Dooremalen and Ellers, 2010). Despite the widespread effects of rearing temperature, regulating temperature in large-scale production systems can be complicated and costly. Furthermore, convective cooling/heating by ventilation systems must be balanced against several other parameters, including a) the metabolic heat production from the larvae and the surrounding microbiomes, b) the prevention of hypercapnia and hypoxia, and c) the need to control humidity to produce larvae and frass (remaining feed and excreta) with desired water content.

Considering both the costs of temperature regulation and the potential effects on growth and metabolism it is important to parametrize to what degree developmental temperature affects growth, body composition, metabolism, and CO₂ production in BSFL under the conditions that prevail in realistic large-scale production systems. To address this question, we conducted this study in two steps. In the first step, we evaluated if a small-scale, laboratory-based respirometry setup (approx. 0.4 kg of feed with 600 BSFL) resulted in comparable estimates of growth rate, metabolic rate, feed conversion ratio, and body composition as was measured in a large-scale commercial production system (>8,000 kg of feed with >12 million BSFL) running at the same experimental temperature. In the second step, we used the laboratory-based system to investigate the effect of rearing temperature (ranging from 27.0 to 41.9°C) to measure survival, larval growth, CO₂ emission, O₂ consumption, respiratory exchange ratio (RER), crude lipid content, crude protein content, and feed conversion ratio of the BSFL. Based on these results, we discuss optimal rearing temperature in terms of productivity, environmental impact, and larval body composition of protein and lipid. Furthermore, the measurements of

the RER are discussed as a potential window to monitor both the growth and macronutrient composition of the produced larvae.

Materials and methods

Experimental animals and feed

For both large-scale and small-scale experiments, we used the black soldier fly larvae (BSFL, *Hermetia illucens*), and feed provided by the BSFL producer ENORM Biofactory A/S, Denmark. The seed larvae were approximately 5-6 days old, corresponding to the 4th or 5th instar (Gligorescu et al., 2019) at the onset of the experiment and larvae were subsequently reared for 7 days at a density of approximately 1.5 seed larvae per gram of wet pre-mixed commercial feed. Feed was a mixture of by-products from the agricultural and food industry (+70%) including side streams from the potato, milk, citrus, and canola industry, combined with grains to give a wet, porridge-like consistency. The composition of the feed is shown in Table 1.

Table 1. Composition of the commercial feed.

Experimental protocols

The present study had two principal research aims: i) Establish whether the small-scale experimental set-up are indicative of outcomes in large-scale production settings, and ii) to investigate the role of rearing temperature on growth, metabolism, and body composition.

Experiment 1: Large-scale production setting

Large-scale growth and metabolism experiments were performed twice under similar conditions in November 2021 and December 2021 at ENORM Biofactory A/S, Denmark. Respectively, 1.080 and 1.320 crates were placed within a 5 x 10.5 x 2.3 m rearing unit (Figure 1). Each plastic crate (60 cm x 40 cm) contained ~7.5 kg of feed and ~11.200 seed larvae with a weight of ~4-13 mg per larvae. Crates were stacked in racks that allowed ventilation of all crates and the automated ventilation system recorded substrate and air temperature (°C) as well as relative humidity (RH, %), airflow (L/min), and CO₂ concentration (%) every 2 minutes from the outgoing air. The ventilation system of the rearing unit was programmed to control airflow, temperature and humidity such that substrate rearing temperature averaged ~34°C with fluctuations confined between 32-38°C, and room air was generally maintained at 70-90% RH over the 7-day growth period.

Figure 1. Schematic overview of the large-scale facility

The concentration of CO₂ (%) in outgoing air was measured with a DOL 19 CO₂ sensor (dol-sensors A/S, Denmark), and for calculations of CO₂ production rate, incoming air was assumed to have a CO₂ concentration of 410 ppm. The average annual CO₂ concentration in 2021 was 413 ppm and atmospheric CO₂ concentrations are generally slightly below average in November and December (URL: <https://www.co2.earth/daily-co2>). Total airflow from the ventilation system was recorded using an in-house calibrated system (SKOV A/S, Denmark). Briefly, a pressure drop was recorded when air flowed across a damper and the recorded pressure drop was then converted to a flow recording using calibrations that considered the position of the damper. Airflow was recalculated to standard dry air by correcting for the temperature and humidity of the outgoing air and assuming a barometric pressure of 760 mmHg.

At the termination of the 7-day growth period, samples were collected from 12 crates (four crates from the front, four from the middle, and four from the back of the room) for later measurements of larvae growth and body composition (Figure 1). Larvae and frass from the

selected crates were weighed and sub-samples were stored at -20°C for subsequent measurements of fractional water, crude lipid, and crude protein content.

Experiment 1: Small-scale laboratory setting

In November 2021, a production crate with ~7.5 kg feed and ~11.200 seed larvae was randomly taken from the production facility at ENORM Biofactory A/S, Denmark, and transported for 1 hour to the research facilities of the Department of Biology, Aarhus University, Denmark. Upon arrival, feed and larvae were mixed and dosed in 7 round 1 L plastic chambers as part of a small-scale respirometry setup (Figure 2). The 7 replicates contained 408 ± 13 g mix of feed and larvae with similar larval density as in the large-scale experiments (~600 larvae per chamber). The 7 respirometry chambers plus an empty control chamber were placed in a 35°C temperature-controlled cabinet and connected to an open-flow respirometry system allowing for hourly measurements of CO₂ production rate and O₂ consumption rate during the subsequent 7-day growth period (see below for details about small-scale respirometry system). The substrate rearing temperature averaged ~35°C with fluctuations confined between 33-38°C. At the termination of the experiment larvae from each replicate were weighed and sub-samples of larvae and frass were stored at -20°C for measurements of fractional water, crude lipid, and crude protein content.

Figure 2. Schematic overview of the small-scale respirometry setup.

Experiment 2: Effects of temperature on growth and metabolism

In the spring of 2022, a series of experiments were performed to investigate the effect of rearing temperature on the growth and metabolism of BSFL using the small-scale respirometry system. For all experimental temperatures, 580-605 larvae (3.4-6.2 mg per larvae) were placed in ~400 g feed in 6 round 1 L respiration chambers. Each replicate chamber contained a data logger (DS1921G Thermochron iButton, Maxim Integrated, USA) that recorded every 30 min to measure the variability and mean temperature. The four treatment temperatures: 27.0, 35.3, 39.4, and 41.9°C represent the average of the substrate temperatures recorded across the 7-day growth period. CO₂ production and O₂ consumption were recorded hourly for 7 days with 2 empty chambers for control measurements. At the termination of the experiment, larvae from each replicate were weighed and samples were stored at -20°C for subsequent measurements of fractional water, crude lipid, and crude protein content.

The small-scale respirometry system

O₂ consumption rate and CO₂ production rate in the small-scale facility were measured using open respirometry in 8 parallel respirometry chambers (diameter 15 cm, height 6 cm, volume 1 L) (Figure 2). During experiments, the respirometry chambers were placed in a temperature-controlled cabinet (Termaks, Norway) programmed to the desired treatment temperature. One or two of the chambers were kept empty to record and control for background concentrations of CO₂ and O₂ and the remaining chambers contained BSFL and feed. Small mesh nets on the chamber in- and output tubes ensured that larvae could not escape. All respirometry chambers were continuously flushed with air through gastight plastic tubes. Airflow through each chamber was controlled using a Flowbar-8 Mass Flow Meter System (Sable Systems, USA) and airflow was adjusted (range 320-900 mL/min) to ensure a stable and equivalent loss of mass as encountered under commercial conditions. Thus, loss of mass (primarily water) in each respirometry chamber was controlled to ~30 g/day by adjusting airflow (drying) or in some cases by adding MilliQ water.

To measure the O₂ consumption and CO₂ production, a sub-sampler pump (SS-4 Sub-Sampler Pump, Sable Systems, USA) with a fixed rate of 200 mL/min was used to sequentially extract sub-samples of air from the outflow tubes of the 8 respirometry chambers. The sampled air passed through a calcium chloride column for removal of water vapor and syringe filters (Ac-rodisc syringe filter, 25 mm, 0.2 µm Supor membrane, Pall Corporation, USA) were placed to prevent particulate matter from entering the gas analysers. An interface module (UI-3 Data Acquisition Interface, Sable Systems, USA) was used to program an 8-channel-multiplexer (RM-8 Respirometry Flow Multiplexer, Sable Systems, USA) to sequentially shift between chambers every 7.5 min resulting in one estimate of CO₂ production rate and O₂ consumption rate from each chamber every hour. Concentrations of O₂ were measured using a differential oxygen analyser (FC-2 Differential oxygen Analyser (Oxzilla), Sable Systems, USA), and concentrations of CO₂ were measured using a CO₂ gas analyser (LI-850 Gas Analyzer, LI-COR Biosciences, United Kingdom). ExpeData software (Sable Systems, USA) was used to control the 8-channel multiplexer and as an AD converter for data acquisition. Respiration measurements represent the metabolism of several hundred larvae and the interacting substrate microorganisms, which makes it unlikely that the metabolic rate will fluctuate dramatically from hour to hour. Accordingly, outlier data were removed, and a quality assessment of data was performed to prevent faulty measurements from blurring the analysis (see Supplementary material, Figure S1 and S2).

Small-scale gas quality assessment, estimation, and calculation method

The measured flow was corrected to an estimated flow of dry air using a minor correction factor of 0.988 to account for the partial pressure constituted by water vapor in atmospheric air assuming that the ingoing room air temperature was at 21°C and 50% RH.

The concentrations of consumed O₂, $\Delta c(\text{O}_2)$, and produced CO₂, $\Delta c(\text{CO}_2)$, were estimated by the difference in concentration between the respiration chambers, $c(\text{O}_2)_{\text{chamber}}$ and $c(\text{CO}_2)_{\text{chamber}}$, and an empty control chamber, $c(\text{O}_2)_{\text{control}}$ and $c(\text{CO}_2)_{\text{control}}$.

$$\Delta c(\text{O}_2) = c(\text{O}_2)_{\text{control}} - c(\text{O}_2)_{\text{chamber}} \quad (1)$$

$$\Delta c(\text{CO}_2) = c(\text{CO}_2)_{\text{chamber}} - c(\text{CO}_2)_{\text{control}} \quad (2)$$

Rates of O₂ consumption, $\dot{V}_{\text{O}_2}$, and CO₂ production, $\dot{V}_{\text{CO}_2}$, (mL/min) were calculated from the flow rate of dry air through the chamber, $\dot{V}_c$, and the concentration difference between replicates and empty control chambers, $\Delta c(\text{O}_2)$ or $\Delta c(\text{CO}_2)$.

$$\dot{V}_{\text{O}_2} = \dot{V}_c \times \Delta c(\text{O}_2) \quad (3)$$

$$\dot{V}_{\text{CO}_2} = \dot{V}_c \times \Delta c(\text{CO}_2) \quad (4)$$

Total O₂ consumption, V_{O_2} , and CO₂ production, V_{CO_2} , (L) for each replicate chamber were calculated for the 7-day growth period by summation of the hourly respiration rate measurement. Quality assessment of the small-scale O₂ and CO₂ data was performed, and > 90% of the original measurements were considered good quality and used for further data analysis (see Supplementary material).

The respiratory exchange ratio (RER) was calculated as $V_{\text{CO}_2}/V_{\text{O}_2}$ and the RER estimate is considered to provide information on the combined impact of metabolic respiratory activity and the surplus CO₂ expelled during lipogenesis from carbohydrates (Talal *et al.*, 2021).

Water, lipid, and protein content

Following the 7-day growth period, larvae and frass of individual crates or respirometry chambers were weighed to assess total mass gain of larvae and to assess total loss of mass and water in the feed. Subsequently we processed subsamples of larvae to assess water, crude lipid, and crude protein content of the larvae.

The water content of either larvae, feed, or frass was determined by measuring the fractional loss of mass following ~24 h of dehydration in an oven set to 70°C. To obtain crude lipid content, sub-samples of dried larvae were weighed and placed in perforated Eppendorf tubes that were then placed inside a Soxhlet apparatus. Here the samples were washed repeatedly with petroleum ether over a 24-hour period to remove the lipid component. The lean samples were then oven dried, and an estimate of the total lipid proportion was calculated from the difference in initial dry mass and lean dry mass. Another sub-set of dried larvae was taken from each of the replicates to assess crude protein content. The dried larvae were ground with a mortar and samples (~5-10 mg) were analysed for nitrogen content using a combustion analyser (Na 2000, Italy). Crude protein content was estimated by multiplying the nitrogen content by a factor of 4.76, which is a standardized conversion factor for protein to nitrogen ratio in BSFL (Janssen *et al.*, 2017). Crude lipid and crude protein content was, for the large-scale experiments, only determined for the second round of experiments.

Data analysis

Data analysis and visualizations were performed in RStudio version 2023.09.1 with R version 4.3.2 (RStudio Team, 2023; R Core Team, 2023). Due to the small sample size with the large-scale setup ($n = 2$), we were unable to statistically test for differences between the large- and small-scale ($n = 7$) experiments.

One-way ANOVA tests were conducted to test for differences between temperature treatments (27.0, 35.3, 39.4, and 41.9°C). The sample size for each small-scale temperature treatment was $n = 6$, and assumptions, including normality and homogeneity of variances were checked and met. Tukey's Honest Significant Difference (HSD) test was used as a post hoc test, and results were considered statistically significant if they met the criterion of $p < 0.05$.

Results

Comparison of growth and metabolism at large- and small-scale

In our initial experiments, we compared growth, metabolism, and CO₂ production rates measured at large-scale production conditions (>8,000 kg, >12 million BSFL, $n = 2$) with values measured under controlled small-scale laboratory conditions (~0.4 kg, ~600 BSFL, $n = 7$). Formal statistical comparison between these systems was not appropriate given that data from the large-scale experiments were only sampled in two independent replicates. Even so, the data qualitatively supports that small-scale laboratory conditions can be considered representative of large-scale production conditions (Figure 3 and Table 2). Measurements of gas exchange from both small- and large-scale systems found the CO₂ production rate to increase slowly in the first 36 hours, after which there was a more potent acceleration in metabolism during the subsequent 48 hours. In both systems, the CO₂ production rate peaked after approximately 72-84 hours (Figure 3). Subsequently, the CO₂ production rate declined steadily in the small-scale chambers during the following 4 days while this decline was less pronounced and more variable under the large-scale conditions. Integrating the total CO₂ production over the 7-day growth period yielded a net CO₂ production of 65.7 ± 1.0 L CO₂ per kg wet mass in the small-scale respiration chambers, while CO₂ production was estimated 20-25% higher for the two independent measuring rounds of CO₂ production at large scale (82.6 and 78.0 L CO₂ per kg wet mass, respectively; Figure 3 and Table 2). Note, however, that measurements of CO₂

production rate for the two systems were performed using different methodologies (see methods and discussion).

Figure 3. Comparison of CO₂ production rates, $\dot{V}\text{CO}_2$, between large- and small-scale facilities.

Table 2. Comparison of production and respiratory parameters for black soldier fly larvae (BSFL) reared in commercial (large-scale) and in laboratory (small-scale) facilities.

Airflow of both small and large growth systems was programmed to gradually dehydrate the frass (mixture of excreta and residual feed) through the removal of water vapor over the 7-day growth period. This is done primarily because a dry texture of frass facilitates efficient separation of larvae and frass during harvest. In our systems, the initial water content of feed ranged between 67.5 and 71.0% and after the 7-day growth period, the water content was reduced to a range of 45.0 and 52.2% in the remaining frass (Table 2). The ventilatory loss of water vapor and CO₂ over the 7-day growth period reduced the total wet mass of larvae and frass to 55-50% of the initial mass in both the small and large system.

Both the large- and small-scale system supported considerable larval growth with limited mortality. Survival was estimated to $86 \pm 16\%$ and $90 \pm 5\%$ for the two rounds of large-scale production with the average size of harvested larvae at 169 and 168 mg wet mass, respectively. In the small-scale system, survival was estimated to $96 \pm 5\%$ with individual larval weighing 163 ± 7 mg. When estimated as the proportion of initial dry mass, the harvest of larval dry mass corresponded to $23.2 \pm 0.9\%$ and $24.7 \pm 2\%$ for the two rounds of large-scale production and $24 \pm 0.7\%$ for the small-scale respirometry chambers (Table 2). Finally, harvested larvae from small- and large-scale systems were of similar body composition with a dry mass of $33.5 \pm 1.7\%$ and $30.7 \pm 1.1\%$ in the two large-scale production rounds and $34.5 \pm 0.7\%$ in the small-scale chambers. The dried larvae from large-scale production contained $33.0 \pm 1.3\%$ crude lipid and $29.8 \pm 0.5\%$ crude protein (only assessed for one round of large-scale production) which was similar to the larvae from small-scale chambers consisting of $33.3 \pm 1.1\%$ crude lipid and $28.4 \pm 0.5\%$ crude protein (Table 2).

Figure 4. The effect of substrate rearing temperature on gas exchange.

Effects of temperature on growth and metabolism of BSFL

Our initial experiment suggests that larval growth and metabolism in small-scale respirometry chambers are representative of large-scale production systems, and we therefore continued to investigate the role of rearing temperature on growth and metabolism in BSFL at 27.0, 35.3, 39.4, and 41.9°C in the small-scale system. As seen from the respiratory data, the dynamic changes in metabolism during the 7-day growth period vary conspicuously with temperature (Figure 4A). All treatments were characterized by sequential increases and decreases in gas exchange that likely reflect how the “average larvae” in the respiration chambers develop through larval instars. However, these transitions occur slower at low temperature (27.0°C) where the final and largest peak in metabolism is reached after 6 days. In contrast, development and metabolism peaked earlier at 35.3 and 39.4°C while the developmental and metabolic transitions were more difficult to pinpoint at 41.9°C. Thus, the highest temperature was most stressful limiting metabolism (Figure 4A). When the total metabolic expenditure during the 7-day growth period was assessed from the total O₂ consumption we found similar metabolic costs at the three lowest rearing temperatures with O₂ consumption of 45.6 ± 2.9 , 45.7 ± 2.4 or 46.2 ± 2.2 L O₂ per kg initial wet mass at 27.0, 35.3 and 39.4°C, respectively. However, metabolism was reduced to less than half at 41.9°C, where the 7-day O₂ consumption was only 22.1 ± 2.5 L O₂ per kg initial mass (Figure 4B). A similar pattern was found for CO₂ production, with the exception that total CO₂ production was marginally, but significantly, lower

at 27.0°C (49.9 ± 1.8 L CO₂ per kg initial wet mass) compared to the two intermediate temperatures (53.7 ± 1.8 and 53.2 ± 1.0 L CO₂ per kg initial mass at 35.3 and 39.4°C, respectively). Again, the slowed metabolism at 41.9°C was reflected in the much lower CO₂ production (23.5 ± 2.4 L CO₂ per kg initial mass) (Figure 4C). The different patterns of O₂ consumption and CO₂ production were also evident from the calculation of the apparent RER. The RER of 1.06 ± 0.03 at 41.9°C was significantly lower compared to 1.10 ± 0.05 , 1.17 ± 0.05 , and 1.15 ± 0.05 at 27.0, 35.3, and 39.4°C, respectively (Figure 4D). Across all treatments, we observed a RER exceeding 1.00, suggesting that CO₂ production stemmed from both catabolic (with RER ranging from 0.7 to 1.0, depending on whether metabolism was fuelled by lipids or carbohydrates) and anabolic processes such as lipogenesis.

Larval survival following the 7-day growth period was similar, and the mean survival rate was above 80% in all temperature treatments (Figure 5A). The three lowest rearing temperatures generated the largest larvae (49.5 ± 3.4 , 54.7 ± 4.1 , and 52.1 ± 2.4 mg dry mass per larvae at 27.0, 35.3, and 39.4°C, respectively) while larvae mass was severely reduced at 41.9°C (17.1 ± 2.6 mg dry mass per larvae) (Figure 5B). These differences were also reflected in the cumulative larval dry mass production per kg initial dry mass of feed and larvae, which decreased significantly with increasing rearing temperature and with total production considerably lower at 41.9°C (24.6 ± 0.5 , 22.4 ± 0.7 , 22.7 ± 0.6 and $8.1 \pm 1.2\%$ at 27.0, 35.3, 39.4 and 41.9°C, respectively) (Figure 5C). Moreover, the emission of CO₂ per g produced dry larvae biomass increased with temperature (Figure 5D).

Figure 5. The effect of substrate rearing temperature on growth and metabolism.

The body composition of larvae reared at different temperatures was not dramatically different, but rather characterized by some minor, but significant, differences (Figure 6). The dry mass content of larvae was significantly lower at 27°C ($33 \pm 0.7\%$) compared to the three other temperature treatments (35.0 ± 1.1 , 35.0 ± 0.2 , and $34.5 \pm 0.2\%$ at 35.3, 39.4, and 41.9°C, respectively) (Figure 6A). The proportion of crude lipid in the dry mass also varied among treatments with the lowest crude lipid content in 27.0°C larvae ($32.7 \pm 1.0\%$) and the highest at 41.9°C ($38.0 \pm 1.1\%$) and intermediate values at the two middle temperatures (36.0 ± 1.5 and $34.2 \pm 1.1\%$ at 35.3°C and 39.4°C, respectively) (Figure 6B). Finally, the proportion of crude protein in the dry larvae was significantly higher in larvae from 27.0°C ($30.7 \pm 0.7\%$) followed by 41.9°C ($29.7 \pm 0.5\%$), 35.3°C ($28.3 \pm 0.3\%$) and 39.4°C ($27.2 \pm 0.4\%$) (Figure 6C).

Figure 6. The effect of substrate rearing temperature on larvae composition.

Discussion

Comparison of growth and metabolism at large- and small-scale

The data from the present study indicated that small-scale laboratory respirometry chambers support gas exchange, growth, and metabolism in BSFL that is comparable, and therefore representative of, conditions in large-scale commercial conditions. This finding is based on our efforts to evaluate the utility and relevance of controlled laboratory studies in relation to large-scale facilities. Specifically, BSFL were reared at similar substrate temperature (~34-35°C) and in similar diets in either small-scale controlled laboratory respiration chambers (~0.4 kg) (Figure 2) or in the larger and more heterogeneous conditions that exist in a rearing unit at ENORM Biofactory A/S (Figure 1) containing >20.000 times more feed and larvae in the system. Despite the massive difference in scale, we found remarkable similarities in growth and energy conversion. Thus, survival, individual larval size, % dry mass, % crude lipid, % crude protein, and final larval mass as a proportion of initial mass all differed by less than 10%

between the two growth systems (Table 2). The results found in the current study do not agree with the findings from previous studies, which found a higher larval weight in big scale (crates of 2,060 cm²) compared to small scale (194 cm²) (Yakti *et al.*, 2022). According to Yakti *et al.* (2022), the difference is related to a higher substrate temperature in larger crates, which may be avoided in the current experiment because of efficient temperature control in both our small and large-scale systems (measured average was ~34-35°C, and fluctuations in substrate temperature ranged only between 32-38°C). Yakti *et al.* (2022) also found that crude protein content is higher and crude fat content is lower for larvae reared at small-scale compared to large-scale while we observed very similar body composition in our small and large systems.

Unlike the similarities in growth data, we estimated CO₂ production per kg initial wet mass to be almost 20% higher in the large-scale system compared to the small-scale system (Figure 3 and Table 2). We speculate that a considerable part of this difference can be explained by the different experimental methodologies used to assess gas exchange in large- and small-scale facilities, respectively. Specifically, CO₂ production measured at large-scale is experimentally inferior to the controlled laboratory conditions because 1) CO₂ concentration in the incoming air is not measured directly (but assumed to be 410 ppm), 2) airflow in the rearing unit is assessed indirectly from a pressure drop across a damper, and lastly, 3) rearing rooms are not completely airtight and the recorded airflow may also be confounded by occasional opening of the entrance door during the 7-day growth period. Considering these methodological differences, a 20% difference is maybe not surprising between the well-controlled small-scale respirometry chambers and the large-scale commercial facilities. Moreover, when CO₂ production is recalculated relative to larvae production (g CO₂/kg produced dry larvae mass) our systems produce 1,632 ± 46 g CO₂/kg dry larval mass for the small-scale systems and 2,164 ± 69 or 2,350 ± 282 g CO₂/kg dry larvae mass for the measurements at large-scale. Both these estimates are well within the range of previous studies (Ermolaev *et al.*, 2019; Pang *et al.*, 2020; Parodi *et al.*, 2020) where CO₂ production from small-scale (>15 kg initial feed) produced between 1,394 to 2,750 g CO₂/kg dry larvae mass (see table 2 in Parodi *et al.*, 2020).

Comparing the large- and small-scale systems qualitatively, it is evident that the temporal development of CO₂ production was initially similar between large- and small-scale systems, particularly during the first two to three days of accelerated growth and metabolism (Figure 3). Conversely, the gradual reduction in CO₂ production rate is more modest and variable in the large-scale system which is likely due to a more heterogeneous growth and development in the >1,000 different crates placed in the rearing unit. This differentiation in development is confirmed by facility staff observations, revealing that crates taken from different positions in the rearing room, may differ in developmental progression on days 4 to 6.

Based on this initial experiment we conclude that the detailed and precise quantitative estimate of CO₂ production rate in our large-scale system was likely confounded by experimental constraints. However, the qualitative information was similar between different months in the large-scale system and thus comparatively informative (i.e. the level of CO₂ production is informative of the intensity of larval growth). Furthermore, tracking the temporal development of CO₂ metabolism remains a powerful method for monitoring metabolism, growth, and development in both small and large systems (see discussion below). Finally, the close similarities in survival, growth, and assimilation between the large- and small-scale systems indicate that small-scale respirometry chambers can be appropriate to investigate production-relevant manipulations of BSFL production. This could include effects of temperature, humidity, diet, texture, and microorganism interactions, that could affect growth efficiency in BSFL (Danieli *et al.*, 2019; Harnden and Tomberlin, 2016; Meng *et al.*, 2023; Raimondi *et al.*, 2020).

Effects of temperature on survival, growth, and metabolism of BSFL

Our study using small-scale laboratory respiration chambers at different temperatures showed that rearing at 27, 35, and 39°C all support efficient growth in BSFL whereas 42°C causes a severe, but sublethal, depression of growth and metabolism. Previous reports of BSFL have found high survival (>80%) in the temperature range of 15-37°C (Chia *et al.*, 2018; Raimondi *et al.*, 2020) while survival at 40°C is ambiguous with both low (Chia *et al.*, 2018) and high (Purkayastha and Sarkar, 2022) estimates reported in the literature. Here we confirm that BSFL survives rearing under production-like conditions at temperatures ranging from 27.0 to 41.9°C while we found 100% mortality at 44.4°C (data from this treatment are not included due to the fatal outcome). In our study design, we initiate the growth experiments using ~5-6 days old seed larvae, and the high survival at 41.9°C could therefore be linked to the shorter exposure duration at this stressful temperature compared to previous studies (Chia *et al.*, 2018). Thus, the consequences of stressful heat exposure are determined both by the intensity (the absolute temperature) and the duration of exposure (Jørgensen *et al.*, 2021). Furthermore, it is possible that later larvae stages are simply more resilient to high temperatures, and it has previously been shown that the susceptibility to heat stress changes considerably between life stages in BSF (Chia *et al.*, 2018; Tomberlin *et al.*, 2009). Despite the high survival at 41.9°C, it is evident that this temperature is stressful, as it causes clear reductions in both growth and metabolism (see below). Production systems with BSFL should thus avoid rearing temperatures above 40°C while the present study suggests that temperatures ranging from 27.0 to 39.4°C all support efficient development and growth of 4th to 7th instar BSFL (Gligorescu *et al.*, 2019). Thus, the metabolic rate at all these temperatures seems to end with a decline that is indicative of a transition to a non-feeding 7th instar larvae (Figure 4). Furthermore, larvae size, and total larvae growth after 7 days are fairly similar between 27.0 to 39.4°C (Figure 5B and 5C) confirming earlier reports describing temperature optima in BSFL (Chia *et al.*, 2018; Raimondi *et al.*, 2020; Harnden and Tomberlin, 2016).

Even if BSFL growth is well supported by a broad temperature range it is important to consider that small differences can be critical to optimize profitability in a large production facility. Optimization relates both to the efficiency of energy upcycling from cheap “low quality” diets towards assimilation in the “high quality” larvae, but also to reducing the duration of the rearing process. Regarding the efficiency of growth, the energy assimilation in larvae is determined by the relative energy allocation towards assimilation (growth) versus metabolism (costs related directly to growth or maintenance) (Laganaro *et al.*, 2021). This proportional energy allocation varies with larvae age and dietary substrate and is also likely to be influenced by temperature and developmental stage (Laganaro *et al.*, 2021; Bekker *et al.*, 2021). In our experimental design we estimated total growth over a fixed period (one week) and our measurements do therefore not allow for a direct quantitative partitioning of the cost of growth or maintenance metabolism during specific growth periods. Even so, the temporal changes in CO₂ production suggest that growth and metabolism are temperature sensitive which may affect the efficiency of the BSFL production in a system where the growth period is fixed. For example, Figure 4A clearly illustrates that the CO₂ production rate fluctuates predictably with three CO₂ peaks for each of the 3 lowest temperature treatments (27.0, 35.3, and 39.4°C). These peaks are likely representative of growth phases in the 4th, 5th, and 6th larval instars and growth phases clearly proceed faster when larvae are reared at higher temperatures. These temporal patterns suggest that most larvae have entered the 7th non-feeding prepupal larval stage by the end of day 5 in the 35.3 and 39.4°C treatments. In contrast, the average larvae grown at 27.0°C likely only transitioned to the 7th instar by the end of day 7 shortly before the termination of the experiment. Although larval instars were not assessed directly in this study, these data suggest that continuous monitoring of metabolic intensity is a potential tool to record larval growth patterns and thereby control (and limit) production time.

Effect of temperature on body composition

BSFL are known to accumulate anhydrous triglyceride (lipid) in the body during the last instars (Arrese and Soulages, 2010; Gligorescu *et al.*, 2019; Talal *et al.*, 2021) and this process may continue even after they have transitioned to the non-feeding stage. Furthermore, it is possible that rearing at different temperatures stimulates some internal biochemical pathways that ultimately affect the larvae body composition in terms of proportional lipid and protein content as has been suggested by findings from mealworms (Bjørge *et al.*, 2018). Considering the experimental design in the present study it is difficult to separate the direct effects of temperature from those of ontogeny (development) since the respiratory trace suggests that larval development has progressed further at 35.3 and 39.4°C than at 27.0°C. Larvae harvested at 27°C are characterized by a lower crude lipid content (32.7%) compared to 35.3 and 39.4°C (36.0 and 34.2%, respectively) suggesting that temperature and/or developmental stage might be a contributing factor besides diet (Nguyen *et al.*, 2015) affecting the macronutrient composition of BSFL. It is possible that larvae at 27.0°C had less time to process the lipogenesis associated with the latest developmental stages compared to the two warmer treatments (35.3 and 39.4°C). This is consistent with the lower RER of 27.0°C larvae (1.10) compared to 35.3 and 39.4°C (1.17 and 1.15, respectively) which are indicative of different levels of total lipogenesis activity. Thus, lipogenesis from carbohydrates results in a net CO₂ production that contributes to the high apparent RER values, and RER values above 1 are common among fast-growing animals (Talal *et al.*, 2021) including BSFL (Parodi *et al.*, 2020).

CO₂ emission

Costs of growth (anabolism) and the basic costs of maintenance of larvae are associated with considerable metabolic turnover, and furthermore, there is likely also a metabolic turnover from the microbial community that lives in the feed and frass (Laganaro *et al.*, 2021). As such it is relevant to consider the CO₂ budget of BSFL production as this represents an important characteristic of this emerging “green” technology (Parodi *et al.*, 2020). In the current study, we show that the total emission of CO₂ per unit of dry larval biomass increased as the temperature increased, showing a significant leap between temperatures of 39.4 and 41.9°C (Figure 5D). We speculate that the higher CO₂ emission per g larvae produced at higher temperatures is linked to a lower net growth efficiency, indicating that more energy is used for maintenance relative to growth. Additionally, as discussed previously, warmer treatments also facilitated faster development so some larvae at 35.3 and 39.4 °C presumably had >2 days where their growth was halted, and the larvae were primarily utilizing stored energy after transitioning to the non-feeding 7th larval stage. In the case of larvae reared at 41.9°C, CO₂ emission per g larvae (Figure 5D) was considerably higher which, in addition to low growth efficiency, could be linked to metabolic costs associated with heat stress such as production of heat shock proteins at the expense of growth (Gonzalez-Tokman *et al.*, 2020). As mentioned previously, the CO₂ emissions measured are not produced from larval metabolism alone but include CO₂ production from microbial metabolism in the feed (Bekker *et al.*, 2021; Parodi *et al.*, 2020). Therefore, it would be relevant to study directly the effects of temperature on microbial metabolism and the contribution of microbial respiration in the overall system.

Conclusion

The data collected in this study qualitatively support that small-scale laboratory facilities are representative of large-scale commercial facilities in terms of survival, growth, and metabolism. The present study also demonstrated that the production of BSFL is efficient across a broad range of temperatures from 27.0 to 39.4°C with respect to survival, larvae size, total larval dry mass, and body composition. Repeated CO₂ measurements during the 7-day period indicated that larval growth was fastest at 35.3 and 39.4°C suggesting that rearing at high tem-

perature could reduce production time. However, the total CO₂ emission per unit of dry larvae biomass increased with temperature, rendering 27.0°C as a more favourable rearing temperature when evaluating larval production in terms of the quantity of larvae biomass per L CO₂ produced over a 7-day period. Further, the larvae from the 27.0°C treatment were characterised by higher crude protein content and lower crude lipid content demonstrating that temperature can affect the macronutrient composition of produced larvae biomass. The optimal temperature therefore depends on the specific goals from the producer (fast growth, lowered CO₂ emission, higher protein content or other key performance indicators) and here we demonstrate how continuous monitoring of gas exchange in production facilities provides valuable information to monitor the dynamics of growth and body composition in BSFL and how it can be a tool to control and limit production time.

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714 **Table 1.** Composition of the commercial feed.

% of wet mass		% of dry mass			
Water	Dry mass	Carbohydrates ¹	Crude protein	Crude fat	Ash
68.5%	31.5%	61.9%	17.0%	3.3%	17.8%

715 ¹Estimated by proximate analysis.

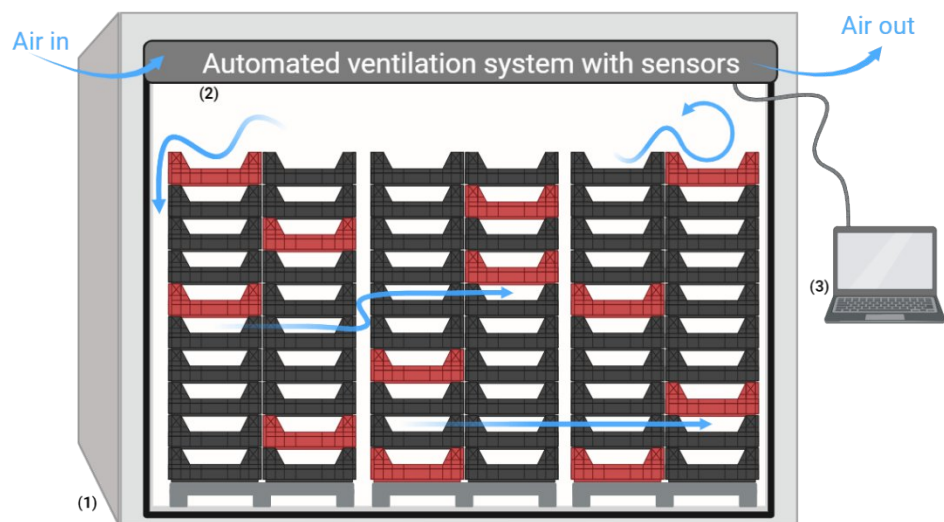


Figure 1. Schematic overview of the large-scale facility. The rearing unit (1) (5 x 10.5 x 2.3 m) contained 1.080 or 1.320 stacked crates (60 x 40 cm) with ~7.5 kg of feed and ~11.200 seed larvae in each crate. The automated ventilation system (2) controlled the climatic conditions through adjustments of the input outside air and the re-circulation to maintain room temperatures between 34-36°C and humidity levels at 70-90% relative humidity. Sensors (2) are installed to record temperature (°C), relative humidity (%), airflow (L/min) and CO₂ concentration (%) that are logged by a computer every 2 minutes (3). The 12 red crates in the figure represent the semi-random sampling approach for technical replicates. Four crates were chosen from the front of the room, four from the middle, and four from the back. Figure created with BioRender.com.

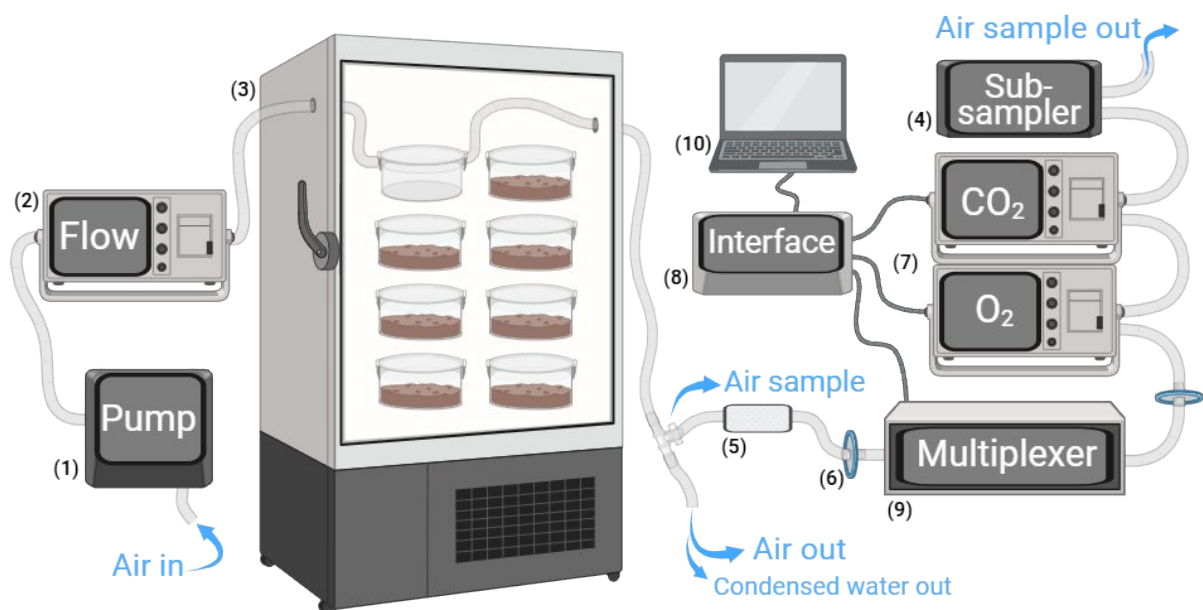


Figure 2. Schematic overview of the small-scale respirometry setup. The flow of air is indicated by the direction of the blue arrows. A pump (1) generates a continuous airflow through each of the 8 chambers (each chamber received individually controlled and continuous airflow, but for simplicity, only one chamber is shown as connected to tubes). The rate of airflow to each chamber is controlled by a multichannel mass flowmeter (2) and the air then passes through 8 respiration chambers (in the figure represented by 1 empty control and 7 replicates containing approx. 400 g of feed and approx. 600 larvae) placed in a temperature-controlled cabinet (3). A sub-sampler (4) extracts sub-samples of air from the tubes leaving the chambers, which passes through a calcium chloride column (5) and a syringe filter (6) before it passes the O₂ and CO₂ gas analysers (7). An interface module (8) controls a multiplexer (9) to sequentially extract sub-samples from the chambers and this resulting data is then recorded on a computer (10) which secures 1 measurement per chamber per hour. Figure created with BioRender.com.

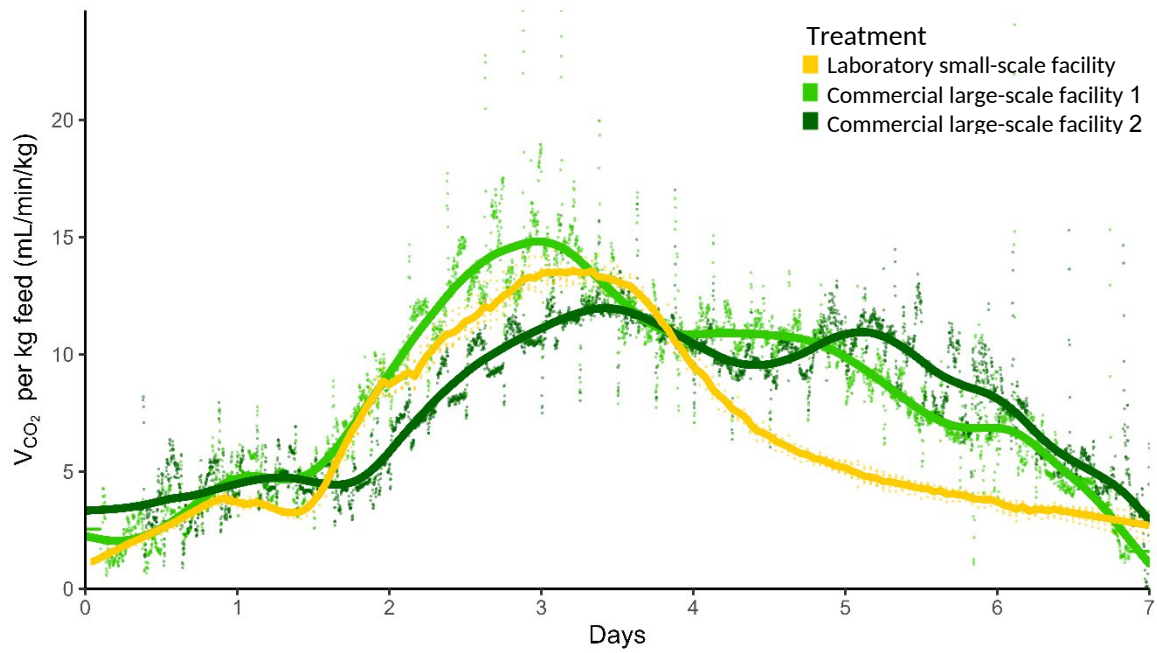


Figure 3. Comparison of CO₂ production rates, $\dot{V}_{CO_2}$, between large- and small-scale facilities. $\dot{V}_{CO_2}$ is reported per kg initial wet mass of feed and larvae and the black soldier fly larvae (BSFL) were reared for 7 days on commercial feed with average substrate temperature of ~34-35°C. Data points are represented as dots. Green lines depict the trend component extracted from the Hodrick-Prescott (HP) filter, with a smoothing parameter set to 1×10^8 to eliminate noise in the two large-scale (>12 million BSFL) experiments. The yellow line shows the mean value obtained from 7 replicates in the small-scale (~600 BSFL) respirometry experiment.

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Table 2. Comparison of production and respiratory parameters for black soldier fly larvae (BSFL) reared in commercial (large-scale) and in laboratory (small-scale) facilities.

Rearing parameters	Commercial facility 1	Commercial facility 2	Laboratory facility
Rearing unit and replicates	Rearing room, 120 m ³ volume, n=1	Rearing room, 120 m ³ volume, n=1	Respiration chambers, 1 L volume, n = 7
Technical replicates from crates	Subsamples from 12 of 1,080 crates, n=12	Subsamples from 12 of 1,320 crates, n=12	1 sample per respiration chamber
Input: feed and larvae			
Initial wet mass of feed and larvae (total kg, average kg per crate)	8,183 kg 7.58 ± 0.05 kg/crate	9,897 kg 7.50 ± 0.09 kg/crate	0.407 ± 0.013 kg
Initial water content of feed (%)	70.3%	71.0%	67.5%
Estimated initial larvae wet mass (total kg, average kg per crate)	124.2 kg 0.12 kg/crate	52.7 kg 0.04 kg/crate	0.003 kg
Estimated no. of larvae per rearing unit (room or chamber)	12 million BSFL	15 million BSFL	600 BSFL
Output: frass and larvae			
Final wet mass of frass and larvae (total kg, average kg per crate)	3,714 kg 3.44 ± 0.26 kg/crate	4,550 kg 3.45 ± 0.11 kg/crate	0.207 ± 0.002 kg
Final water content of frass (%)	45.0%	45.9%	52.2%
Estimated larvae survival (%)	86 ± 16%	90 ± 5%	96 ± 5%
Final larvae wet mass (total kg, average kg per crate)	1,733 kg 1.60 kg/crate	2,218 kg 1.68 kg/crate	0.092 ± 0.004 kg
Individual larvae size (mg wet mass)	169 mg	168 mg	163 ± 7 mg
Larvae DM produced relative to initial DM of feed and larvae (%)	23.2 ± 0.9%	24.7 ± 2%	24 ± 0.7%
Larvae composition			
DM content (%)	33.5 ± 1.7%	30.7 ± 1.1%	34.5 ± 0.7%
Crude lipid content (% of DM)	NA	33.0 ± 1.3%	33.3 ± 1.1%
Crude protein content (% of DM)	NA	29.8 ± 0.5%	28.4 ± 0.5%
Respiration data			
Total CO ₂ production per kg initial wet mass of feed and larvae	82.6 L/kg	78.0 L/kg	65.7 ± 1.0 L/kg
RER (CO ₂ production/O ₂ consumption)	NA	NA	1.21 ± 0.02

Experiments were performed with 5-6 days old black soldier fly larvae (BSFL) reared on pre-mixed commercial feed for 7-days with average substrate temperature of ~34-35°C. The two commercial facility experiments were conducted at ENORM Biofactory, Denmark. The laboratory facility experiment was conducted at Aarhus University, Denmark.

Values indicate mean ± SD, where possible; SD = standard deviation; n = number of replicates or number of technical replicates; DM = dry mass; RER = respiratory exchange ratio; NA = not applicable.

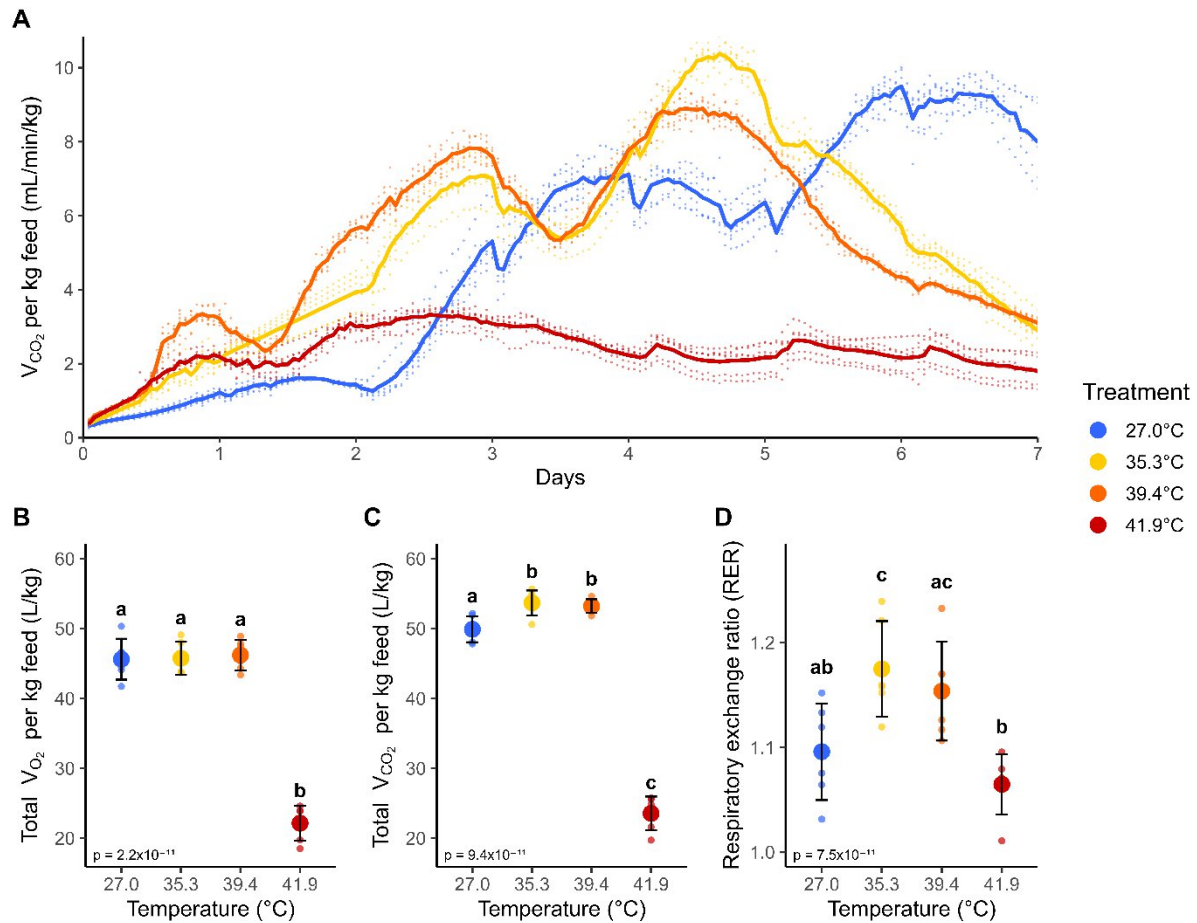


Figure 4. The effect of substrate rearing temperature on gas exchange. Black soldier fly larvae (BSFL) were reared for 7 days on commercial feed within a small-scale setup with 6 replicates. (A) Shows temporal variation in CO_2 production rates, $\dot{V}_{CO_2}$, per kg initial wet mass (feed and larvae). (B) and (C) shows the total volume of O_2 consumed (V_{O_2}) and CO_2 produced (V_{CO_2}) per kg initial wet mass of substrate and larvae in the system, respectively and (D) depicts the respiratory exchange ratio (RER) calculated as V_{CO_2}/V_{O_2} . Replicate values are shown by smaller points, and treatment means are depicted by either a line or a large point. Error bars denote standard deviations, and one-way ANOVA p-values are displayed for each parameter. Different letters indicate significant treatment group differences (Tukey HSD test, $p < 0.05$, $n = 6$).

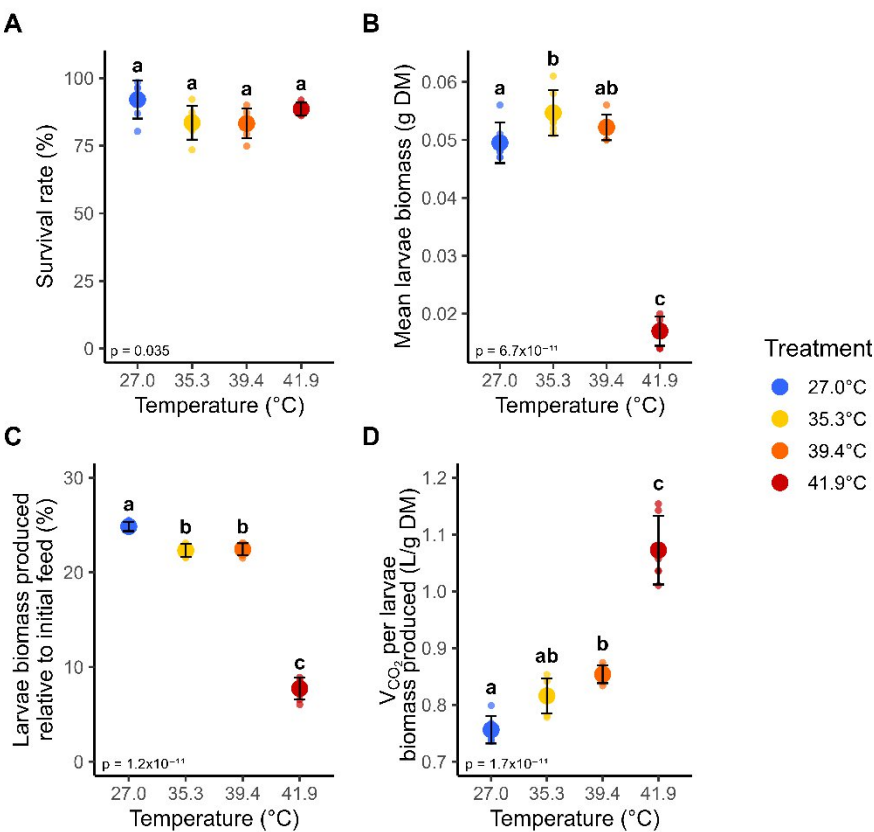


Figure 5. The effect of substrate rearing temperature on growth and metabolism. Black soldier fly larvae (BSFL) were reared for 7 days on commercial feed within a small-scale setup with 6 replicates. (A) Show the survival rate. (B) Show the mean dry mass of individual larvae at the termination of the experiment. (C) Illustrates the proportion of produced larvae dry mass relative to the initial dry mass input of feed and larvae. (D) Displays the volume of CO₂ produced, V_{CO₂}, per g dry mass larvae produced. Replicate values are shown by smaller points, and treatment means are depicted by a large point. Error bars denote standard deviations, and one-way ANOVA p-values are displayed for each parameter. Different letters indicate significant treatment group differences (Tukey HSD test, p < 0.05, n = 6). DM = dry mass.

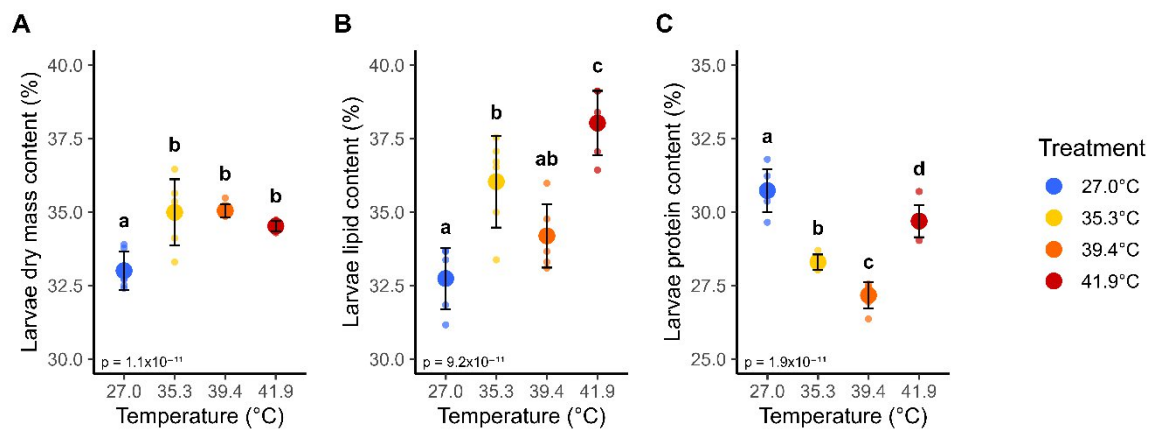


Figure 6. The effect of substrate rearing temperature on larvae composition. Black soldier fly larvae (BSFL) were reared for 7 days on commercial feed within a small-scale setup with 6 replicates. The plots show the (A) proportion of larvae dry matter, (B) the proportion of crude lipid in the dried larvae, and (C) the proportion of crude protein in dried larvae. Replicate values are shown by smaller points, and treatment means are depicted by a large point. Error bars denote standard deviations, and one-way ANOVA p-values are displayed for each parameter. Different letters indicate significant treatment group differences (Tukey HSD test, $p < 0.05$, $n = 6$). DM = dry mass.